Modeling Airline Arrival Delays: A Linear Regression Approach

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Abstract

This report investigates airline arrival delays using the Airline-On-Time Performance dataset from the U.S. Bureau of Transportation Statistics. The analysis focuses on flights in 2019 for selected major airports and carriers. Linear regression techniques are applied to explore how different operational delay components contribute to total arrival delays. Model selection, diagnostic checks, and simulation are performed to assess the statistical properties of the fitted model. A key limitation of the analysis is that total arrival delay is constructed as the sum of several delay components, which creates an almost deterministic relationship between the response and some predictors.

1 Introduction

Airline arrival delays represent a significant operational challenge in modern air transportation systems, affecting both airlines and passengers. Understanding the factors that contribute to these delays is essential for improving scheduling efficiency and overall service quality.
Linear regression techniques are employed to model the relationship between arrival delays and several explanatory variables, including weather-related delays, carrier delays, and air traffic system delays. The analysis is supported by exploratory data analysis, variable selection procedures, and diagnostic checks to evaluate the statistical properties of the fitted model.
A key aspect of this analysis is that the response variable is constructed as the sum of several delay components. This results in a near-deterministic relationship between the response and some predictors, which has important implications for model interpretation.

The remainder of the report is organized as follows. Section 2 presents the objectives of the study. Section 3 describes the dataset and preprocessing steps. Subsequent sections illustrate the exploratory analysis, model selection process, and diagnostic evaluation of the final model.

2 Objectives

The main objectives of this study are:

- to investigate the relationship between airline arrival delays and different operational delay components;
- to develop a linear regression model to explain variations in total arrival delays;
- to apply variable selection techniques in order to identify the most relevant predictors;
- to assess the adequacy of the fitted model through diagnostic analysis;
- to critically discuss the limitations arising from the use of aggregated operational data.

3 Data description

The dataset used in this study is the *Airline On-Time Statistics and Delay Causes*, provided by the Bureau of Transportation Statistics (available at `https://transtats.bts.gov/OT_Delay/OT_DelayCause1.asp`). It contains detailed information on airline operational performance, including flight delays and their causes, for commercial carriers and airports in the United States.
The full dataset spans the period from June 2003 to January 2025 and includes 398,233 observations and 21 variables.
In this study, the response variable is `arr_delay`, representing the total number of arrival delay minutes accumulated within each observation. Negative values correspond to early arrivals.
It is important to note that the dataset is reported at an **aggregated level**. Each observation corresponds to a group of flights defined by a specific combination of carrier, airport, and month. Consequently, the delay variables represent the total delay minutes accumulated by all flights in that group rather than the delay of an individual flight.
The dataset includes several operational variables describing different sources of delay, such as carrier-related delays, weather delays, and delays caused by the National Aviation System (NAS).

3.1 Data preprocessing

Since the dataset described in the previous section is large and highly heterogeneous,a preprocessing step is required before proceeding with the regression analysis.
The full dataset spans multiple years, airports, and airline carriers, which may introduce substantial variability and make the interpretation of the results less transparent. To obtain a more homogeneous and analytically manageable sample, the analysis is restricted to the following subset of observations:

- flights operated in **2019**,
- **four major U.S. airports**: JFK, LAX, ORD, ATL,
- **three airline carriers**: AA, DL, UA.

This restriction reduces the heterogeneity of the dataset while preserving sufficient variability for meaningful statistical analysis.
However, the results should be interpreted within the context of the selected airports and carriers and therefore cannot be directly generalized to the entire U.S. aviation system.
Given the aggregated structure of the dataset, the response variable is rescaled by dividing the total arrival delay minutes (`arr_delay`) by 10,000. This transformation improves numerical stability and readability in regression analysis without affecting the interpretation of the relationships between variables.
Several variables in the dataset are categorical in nature. In particular, `month`, `carrier`, and `airport` are converted into factors in order to properly account for their qualitative nature in both the exploratory analysis and the regression modeling.
Finally, the dataset contains pairs of variables conveying the same information, namely `airport` and `airport_name`, as well as `carrier` and `carrier_name`. Including both variables in each pair would lead to perfect multicollinearity. For this reason the descriptive label variables are excluded from the analysis, while their coded counterparts are retained for modeling purposes.

4 Exploratory analysis

Before fitting the regression model, an exploratory data analysis (EDA) was conducted to better understand the structure of the dataset and the relationships between the response variable and potential predictors.
Figure 1 compares the distribution of the response variable with the distribution of the residuals obtained from the fitted full model. The response variable shows a right-skewed distribution, where most observations correspond to relatively small delays while a few larger delays appear in the right tail. This pattern is typical for delay data, where occasional disruptions may produce substantially larger delays.
In contrast, the distribution of the residuals appears more symmetric and centered around zero. The slight deviation on the right side is mainly driven by a small number of extreme observations rather than by a systematic skewness pattern. Overall, the residual distribution does not indicate substantial departures from the normality assumption required for linear regression.

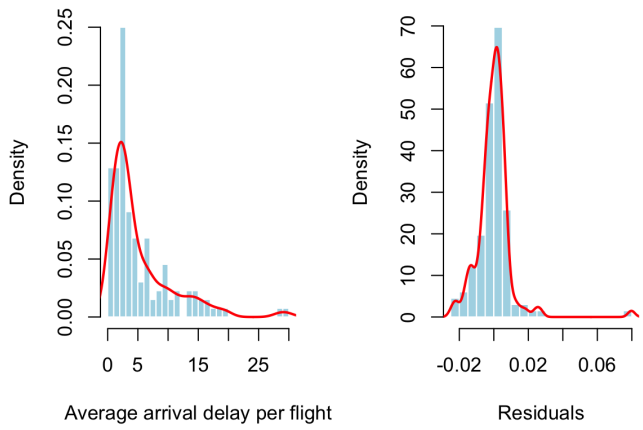


Figure 1: Distribution of the response variable and distribution of the model residuals.

The scatterplot in Figure 2 illustrates the relationship between weather-related delay and the total arrival delay. A clear positive association can be observed: as the number of minutes of weather delay increases, the overall arrival delay also tends to increase. This relationship is expected since weather disruptions represent one of the main operational sources contributing to total delays. The fitted regression line highlights this positive trend. A few observations with particularly large weather delays appear in the upper-right portion of the plot, reflecting periods of unusually severe weather-related disruptions.

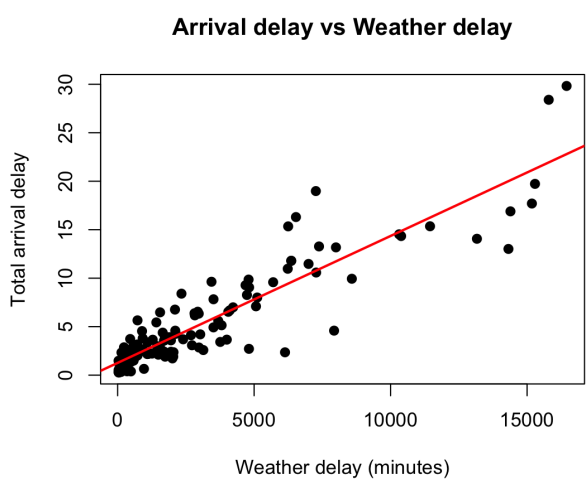
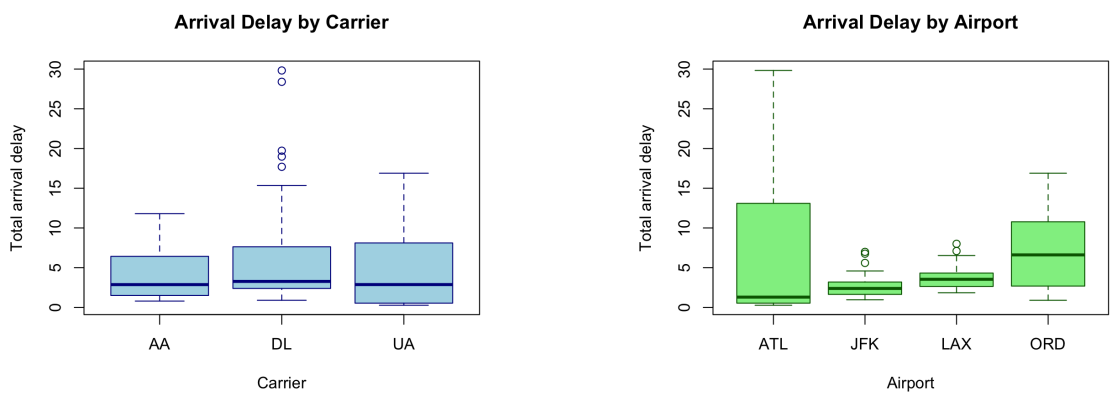


Figure 2: Relationship between weather-related delay and total arrival delay (scaled).

The boxplots below compare the distribution of total arrival delays across airline carriers and airports.
The left panel illustrates the variability of delays across the three selected carriers. The median delay levels appear relatively similar among the airlines, although Delta Airlines (DL) exhibits a slightly higher median and several extreme observations corresponding to unusually large delays.
The right panel shows the distribution of delays across the selected airports. Differences between airports are more pronounced. In particular, Chicago O'Hare International Airport (ORD) displays higher median delays compared to the other airports, while Hartsfield–Jackson Atlanta International Airport (ATL) shows a larger variability with several extreme observations. These patterns suggest that airport-specific operational conditions may play an important role in explaining variation in arrival delays.



(a) Arrival delay by carrier

(b) Arrival delay by airport

Figure 3: Distribution of arrival delays across airline carriers and airports.

The correlation matrix below summarizes the relationships among the main numerical variables included in the analysis.
Figure 4 presents the correlation matrix of the main numerical variables.
Strong positive correlations can be observed between total arrival delay and different delay components, including weather delay, carrier delay, NAS delay, and late aircraft delay. This pattern is expected since these variables represent different operational sources that contribute to the overall delays in arrival.
In addition, relatively high correlations can also be observed among several delay components, suggesting potential multicollinearity among predictors. This issue will be further considered during the model selection phase.

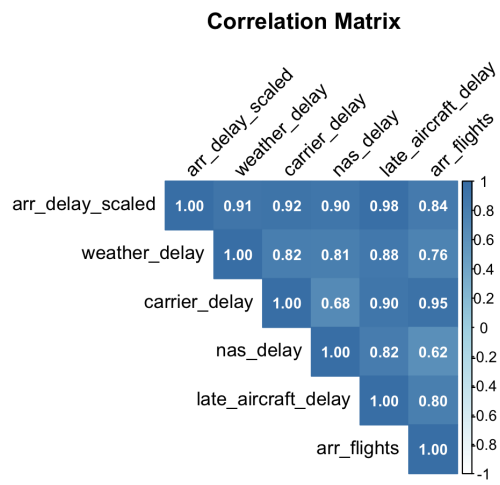


Figure 4: Correlation matrix of the main numerical variables.

5 Variable selection

5.1 Full model

To identify a parsimonious model explaining variation in arrival delays, we start from a full linear regression model.

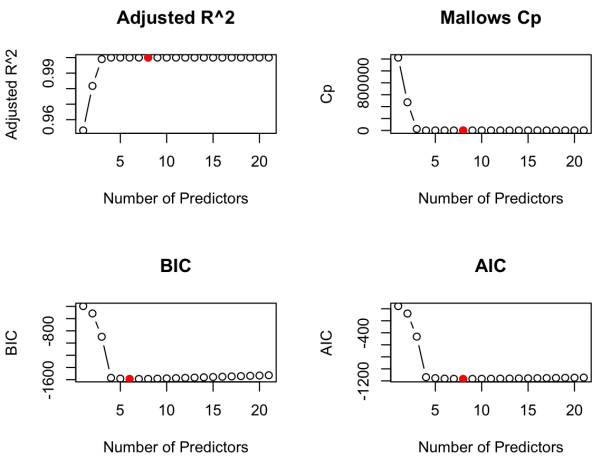
```
# Fit the full linear regression model
full_model <- lm(arr_delay_scaled ~
                 weather_delay + carrier_delay + nas_delay +
                 late_aircraft_delay + arr_flights +
                 airport + carrier + month,
                 data = Airline_sub)
```

5.2 Best subset selection algorithm

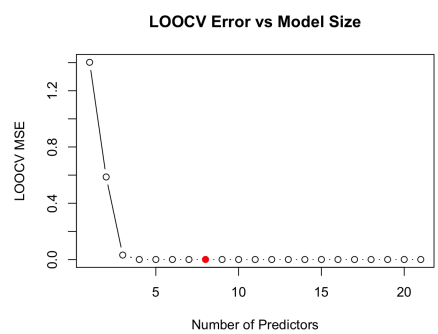
A best subset selection procedure is then applied to explore different combination of predictors and identify the models that provide the best trade-off between goodness of fit and model complexity.

```
X_full <- model.matrix(full_form, data = Airline_sub)
pmax   <- ncol(X_full) - 1

best_subset <- regsubsets(full_form,
                          data  = Airline_sub,
                          nvmax = pmax)

sum_bs <- summary(best_subset)
```

```r
set.seed(1)

for (j in 1:k) {
  best.fit <- regsubsets(full_form,
                         nvmax = p,
                         data = Airline_sub[folds != j,])
}
```



(a) Information criteria for model selection.



(b) LOOCV error as a function of model size.

Figure 5: Model selection criteria used to determine the optimal number of predictors.

The plots above summarize the results of the best subset selection procedure according to several model selection criteria. In particular, AIC, adjusted R^2, and LOOCV suggest selecting a model with 8 predictors, while BIC selects a slightly smaller model with six predictors.

This discrepancy is expected since the criteria penalize model complexity differently. BIC applies a stronger penalty for additional parameters, while AIC and cross-validation tend to favor slightly larger models when the improvement in fit is small.
In this case, the differences in the criteria values between models with 6 and 8 predictors are relatively small. According to the principle of parsimony, when models provide comparable explanatory power, the simpler specification is generally preferred.
Therefore, the model with six predictors selected by BIC is retained for the subsequent diagnostic analysis and inference.

5.3 Final model

The final regression model can be written as:

$$\begin{aligned}\text{arr_delay_scaled}_i = \beta_0 &+ \beta_1\,\text{weather_delay}_i + \beta_2\,\text{carrier_delay}_i + \beta_3\,\text{nas_delay}_i \\ &+ \beta_4\,\text{late_aircraft_delay}_i + \beta_5\,\mathbb{1}(\text{airport}_i = \text{LAX}) \\ &+ \beta_6\,\mathbb{1}(\text{carrier}_i = \text{DL}) + \beta_7\,\mathbb{1}(\text{carrier}_i = \text{UA}) \\ &+ \beta_8\,\mathbb{1}(\text{month}_i = \text{June}) + \varepsilon_i\end{aligned}$$

where ε_i represents the random error term.
All delay-related variables `weather_delay`, `carrier_delay`, `NAS_delay` and `late_aircraft_delay` have positive and highly significant coefficients, indicating that increases in these delay components are associated with larger arrival delays.
Among the categorical variables, flights departing from LAX show a positive effect on arrival delays compared with the baseline airport. Conversely, the coefficients associated with carriers DL and UA are negative, suggesting slightly lower delays relative to the reference carrier.

Finally, the coefficient for month 6 (June) is positive and statistically significant, indicating higher average delays during June relative to the baseline month.

5.4 Collinearity issue

Multicollinearity among predictors was assessed using the Variance Inflation Factor (VIF). As a rule of thumb, VIF values above 10 indicate potentially problematic multicollinearity.
Table 1 reports the VIF values for the predictors included in the final model.

Predictor	VIF
weather_delay	5.36
carrier_delay	9.64
nas_delay	4.36
late_aircraft_delay	15.30
airportLAX	1.14
carrierDL	1.73
carrierUA	1.46
month6	1.07

Table 1: Variance Inflation Factors for predictors in the final model

Categorical variables show very low VIF values, indicating no collinearity issue.
Some of the delay components instead exhibit higher VIF values, with the largest value observed for `late_aircraft_delay`. This is expected because these variables represent different components contributing to the total arrival delay and are therefore naturally correlated.
Despite this moderate multicollinearity, all predictors were retained since they represent distinct operational causes of delay and remain highly statistically significant in the model.

6 Diagnostics

6.1 Heteroscedasticity assumption

To assess the assumption of constant variance, we examined the residuals versus fitted values plot.

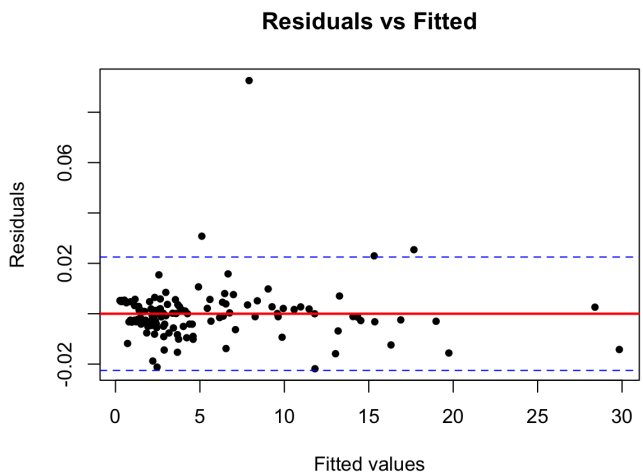


Figure 6: Residuals versus fitted values for the final linear regression model.

Figure 6 displays the residuals of the fitted linear regression model against the predicted values of the response variable `arr_delay_scaled`, defined as the total arrival delay divided by 10,000.
The residuals appear to be randomly scattered around zero with no clear funnel shape or systematic pattern. The spread of the residuals remains relatively stable across the range of fitted values, suggesting that the assumption of homoscedasticity is reasonably satisfied.
Although a few observations exhibit slightly larger residuals, these correspond to extreme delay events and do not indicate a systematic increase in variance.

6.2 Linearity Assumption

To assess the linearity assumption, residual plots were examined against the main numerical predictors included in the model.
In most cases, the residuals appear randomly scattered around zero with no clear systematic curvature, suggesting that the linear specification of the model is appropriate. The absence of strong patterns indicates that the linear relationship between the predictors and the response variable provides a reasonable approximation of the underlying data structure.

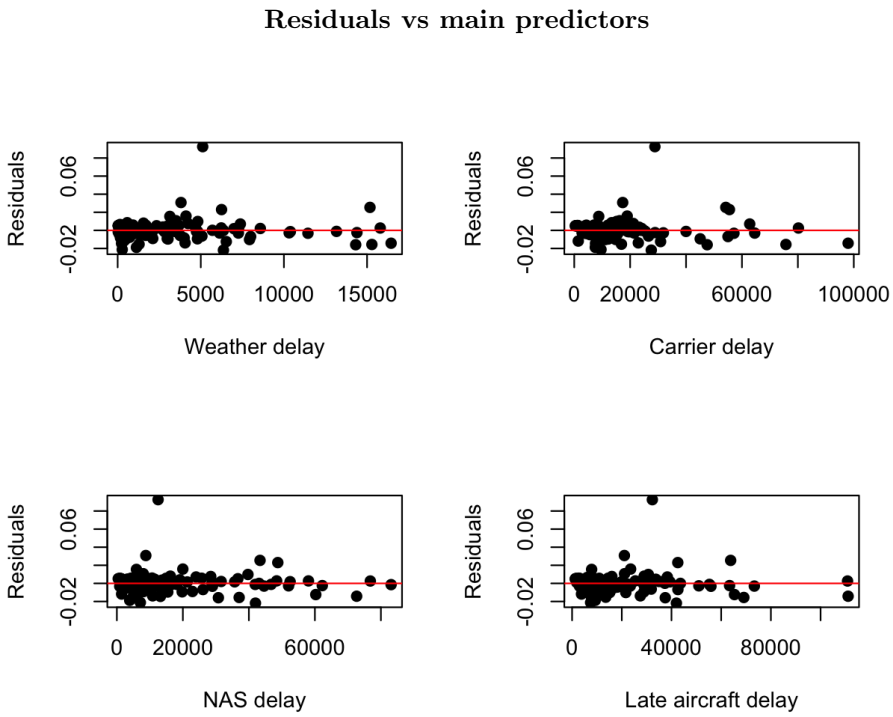


Figure 7: Residuals plotted against the continuous predictors (weather delay, carrier delay, NAS delay, and late aircraft delay) for the final linear regression model.

6.3 Normality assumption

The Q–Q plot indicates that the residuals closely follow the reference line in the central part of the distribution, suggesting approximate normality for the majority of observations.
Some deviations appear in the tails of the distribution, particularly in the upper tail where a small number of observations exhibit larger residuals. This behaviour is mainly driven by a limited number of extreme delay events.
Although the Shapiro–Wilk test rejects the null hypothesis of normality, this result is not unexpected given the presence of a few extreme observations. Overall, the graphical diagnostics indicate that the residuals are reasonably close to a normal distribution for the purposes of linear regression inference.

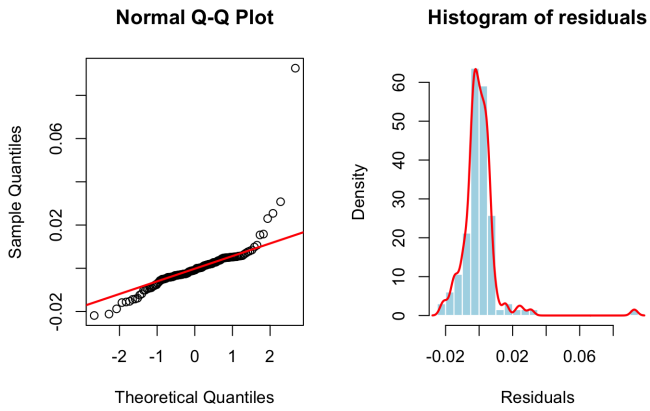
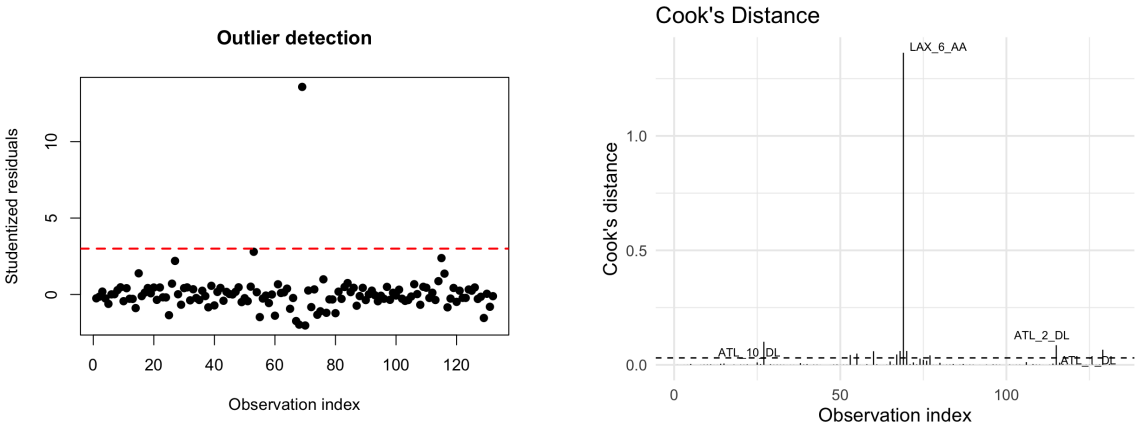


Figure 8: Normal Q–Q plot and histogram of the residuals of the final regression model

6.4 Outlier and influential observation



(a) Studentized residuals used for detecting potential outliers.

(b) Cook's distance used for detecting influential observations.

Figure 9: Detection of outliers and influential observations in the fitted regression model.

Outliers were first assessed using studentized residuals, where observations with $|r_i| > 3$ are commonly considered extreme. As shown in Figure 9, only one observation exceeds this threshold, indicating a substantial discrepancy between the observed and predicted values. In particular, the flagged observation corresponds to the combination `LAX--AA` in month 6 (June) and shows a studentized residual of $r_{69} = 13.58$. Inspecting its delay components reveals unusually large aggregated values (e.g. late aircraft delay = 32,299 minutes, carrier delay = 28,863 minutes, NAS delay = 12,492 minutes), which is consistent with a rare disruption event. Since the response variable is

defined as `arr_delay_scaled` = `arr_delay`/10000, extreme aggregated delays naturally translate into large residuals.

To further assess whether this observation has a strong impact on the estimated regression coefficients, Cook's distance was computed. While the same observation exhibits a relatively large Cook's distance, most observations remain well below the commonly used thresholds (e.g. $4/n$ or 1).
Overall, the results suggest that the presence of a small number of extreme observations does not substantially compromise the stability of the fitted regression model.

7 Interpretation of the coefficients and their uncertainties

The coefficients of the final models are reported in Table 2

Predictor	Estimate	Std. Error	t-value	p-value
Intercept	0.00211	0.00204	1.033	0.304
weather_delay	0.00010095	0.00000062	161.830	< 0.001
carrier_delay	0.00010041	0.00000019	539.948	< 0.001
nas_delay	0.00009995	0.00000012	827.827	< 0.001
late_aircraft_delay	0.00009976	0.00000020	500.418	< 0.001
airport (LAX)	0.00855	0.00242	3.527	< 0.001
carrier (DL)	-0.00560	0.00277	-2.021	0.045
carrier (UA)	-0.00726	0.00274	-2.648	0.009
month (June)	0.01698	0.00379	4.482	< 0.001

Table 2: Estimated coefficients of the selected linear regression model explaining scaled arrival delays.

The coefficients associated with the delay components (`weather_delay`, `carrier_delay`, `NAS_delay`, and `late_aircraft_delay`) are all positive and highly significant. This indicates that increases in these aggregated delay components are associated with increases in the total arrival delay. Since both the response variable and the predictors are measured in minutes (with the response scaled by 10,000), the coefficients can be interpreted as the proportional contribution of each delay source to the overall arrival delay.
Among the categorical predictors, flights departing from LAX show a positive coefficient relative to the baseline airport, indicating higher average arrival delays for this airport when holding other factors constant.
Conversely, the coefficients for carriers DL and UA are negative, suggesting that these carriers tend to experience slightly lower arrival delays compared to the reference carrier.

Finally, the coefficient associated with month 6 (June) is positive and statistically significant, indicating that average arrival delays tend to be higher in June relative to the baseline month

7.1 Individual significance tests for the regression coefficients

For each coefficient of the final linear regression model, the null hypothesis is tested :

$$H_0 : \beta_j = 0 \qquad \text{against} \qquad H_1 : \beta_j \neq 0,$$

using the two-sided t-test. The test is given by:

$$t_j = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)},$$

which, under the null hypothesis, follows a t distribution with the appropriate residual degrees of freedom.

Table 2 reports the estimated coefficients, their standard errors, t-values and associated p-values. Overall, all delay-related predictors are highly statistically significant, while the intercept is not significant at the 5% level.
Among the categorical predictors, the coefficient associated with airport LAX is positive and highly significant, whereas the coefficients for carriers DL and UA are negative and statistically significant. Finally, the indicator for month 6 (June) is positive and statistically significant.

7.2 Testing a group of predictors

To assess the joint relevance of groups of regressors, we performed F-tests comparing the full model with appropriate reduced models.
First, we tested the four delay-related predictors `weather_delay`, `carrier_delay`, `nas_delay`, and `late_aircraft_delay` as a group. These variables represent the main operational sources of flight delays and therefore form a natural block of regressors.
The null hypothesis tested was

$$H_0 : \beta_{\text{weather}} = \beta_{\text{carrier}} = \beta_{\text{nas}} = \beta_{\text{late aircraft}} = 0,$$

against the alternative that at least one of these coefficients differs from zero.
The comparison between the reduced model and the full model yields an extremely large F statistic (approximately 6.5×10^6) and a p-value smaller than 2.2×10^{-16}. Therefore, the null hypothesis is strongly rejected. This indicates that the delay-related predictors are jointly highly significant.

We also considered the global F-test for the entire regression model, testing

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_8 = 0.$$

The resulting F statistic is approximately 3.57×10^6 with a p-value smaller than 2.2×10^{-16}, leading to a clear rejection of the null hypothesis. Hence, the regression model as a whole is highly statistically significant.

Since the response variable is defined as `arr_delay_scaled` = `arr_delay`/10000, these results imply that both the aggregated delay components and the categorical indicators jointly explain a substantial portion of the variation in the scaled total arrival delay.

7.3 Goodness of fit

The goodness of fit of the final regression model can be evaluated using standard measures such as the coefficient of determination (R^2),the adjusted R^2, and the residual standard error.
From the model summary, both R^2 and the adjusted R^2 are equal to 1, indicating that the model explains nearly all the variability of the response variable.

This result should be interpreted in light of the structure of the dataset. The response variable is defined as

$$\texttt{arr_delay_scaled} = \frac{\texttt{arr_delay}}{10000},$$

where `arr_delay` represents the total aggregated arrival delay.
Moreover, each statistical unit in the dataset corresponds to a specific combination of airport, carrier, and month. Therefore, the response variable measures the aggregated arrival delay for each airport–carrier–month combination.
Since several predictors in the model represent the main components of flight delays (weather delay, carrier delay, NAS delay, and late aircraft delay), it is expected that they explain most of the variability in the total aggregated delay.
The residual standard error of the model is approximately 0.0116 on the scaled response. When translated back to the original scale, this corresponds to about $0.0116 \times 10000 \approx 116$ minutes of aggregated delay.
Given the magnitude of the aggregated delays observed in the dataset, this indicates that the model provides a very accurate fit to the data.

8 Prediction

In order to evaluate the predictive capability of the fitted regression model, we consider a hypothetical new observation of the regressors.
Rather than selecting arbitrary values, we construct a realistic scenario by setting the quantitative delay components equal to their sample mean values observed in the dataset. This choice ensures that the new observation represents typical delay conditions in the data.

```r
#Prediction
colMeans(Airline_sub[,c("weather_delay",
                        "carrier_delay",
                        "nas_delay",
                        "late_aircraft_delay")])
new_obs <- data.frame(
  weather_delay = 3049.485,
  carrier_delay = 15661.992,
  nas_delay = 15474.758,
  late_aircraft_delay = 18322.742,
  airportLAX = 1,
  carrierDL = 0,
  carrierUA = 0,
  month6 = 1
)
predict(final_model, newdata = new_obs)
```

Recall that each statistical unit in the dataset corresponds to a specific combination of airport, carrier, and month.

In this example we consider the case of flights departing from LAX, operated by the baseline carrier (American Airlines), during June. The delay components are set equal to the average values observed in the sample.

Using the fitted regression model the predicted value of the response variable for this new observation is

$$\widehat{\text{arr_delay_scaled}} = 5.28281.$$

Since the response variable is defined as

$$\texttt{arr_delay_scaled} = \frac{\texttt{arr_delay}}{10000},$$

this corresponds to an estimated aggregated arrival delay of approximately

$$5.28281 \times 10000 \approx 52\,828 \text{ minutes.}$$

Therefore, under typical delay conditions and for the airport–carrier–month combination considered (LAX, American Airlines, June), the model predicts a total aggregated arrival delay of about 52,800 minutes.
This prediction refers to the aggregated delay associated with the specific airport–carrier–month statistical unit, rather than to the delay of a single flight.

9 Simulation

The simulated responses were generated using the estimated regression coefficients as the true parameters and adding random noise based on the estimated residual standard deviation of the fitted model.

```
set.seed(12)

beta = coefficients(final_model)
X = model.matrix(final_model)
n = nrow(X)

y_hat = X %*% beta + rnorm(n, 0, sigma(final_model))
y_obs = final_model$model[,1]
```

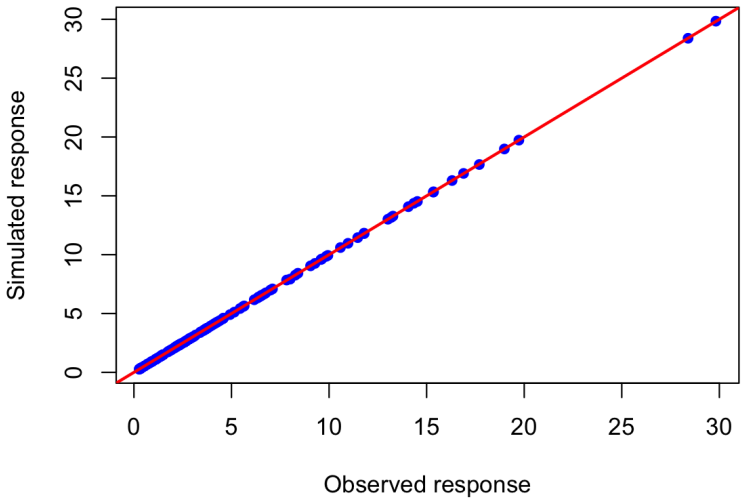


Figure 10: Comparison between the observed and simulated values of the response variable

Figure 10 compares the observed and simulated values of `arr_delay_scaled`. The simulated values lie very close to the observed ones, producing points concentrated around the diagonal.
This behavior is expected in the present dataset because the response variable represents aggregated arrival delay and is strongly related to the delay components included as predictors (weather, carrier, NAS and late aircraft delays). Since these predictors account for most of the variation in the aggregated delay, the fitted model explains almost all variability in the response.

Consequently, the residual variance is very small and the simulated responses are nearly identical to the observed values, indicating that the regression model closely reproduces the structure of the aggregated delay data.

10 Conclusion

This study analyzed airline arrival delays using linear regression techniques applied to aggregated operational data. The results show that several delay components, such as weather and carrier delays, play a significant role in explaining variations in total arrival delays. The model demonstrates a high explanatory power; however, this result must be interpreted with caution. Since the response variable is constructed as the sum of several delay components, a near-deterministic relationship is introduced between the response and some predictors. Furthermore, the aggregated nature of the dataset reduces variability and limits the interpretation of the results at the individual flight level. Therefore, the model should be understood as describing structural relationships within the data rather than providing strong predictive performance.

Overall, this analysis highlights both the usefulness and the limitations of linear regression models when applied to aggregated operational datasets.